Tianlan Wei

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Education

University of Maryland 01/2025 – Present

Doctor of Philosophy program in Geographical Sciences

Johns Hopkins University 08/2023 – 08/2024

Master of Science in Information Systems

Zhejiang University of Technology 09/2019 – 06/2023

Bachelor of Management in Business Administration

Waterford Institute of Technology 09/2021 – 12/2021

International Exchange Program

Professional & Research Experience

Department of Geographical Science, University of Maryland

Research Assistant 01/2025 – Present

Faculty Specialist 09/2024 – 01/2025

- **Breakeven Analysis**: Conducted a breakeven analysis using GCAM to determine the critical methane leakage thresholds and transport distances, evaluating the environmental competitiveness of US LNG against coal in European and Asian markets.
- **Market Scenario Modeling**: Simulated global LNG market dynamics under cost variation scenarios (High vs. Low Cost), projecting future demand trajectories and trade flows through 2050 to assess market resilience.
- **Historical Emission Drivers**: Applied LMDI decomposition to analyze the drivers of LNG imports in South Korea, India, and the UK, quantifying the historical emission reductions achieved by coal-to-gas switching strategies.
- **Ongoing Research**: Drafting a paper on global gas trade restructuring, utilizing counterfactual analysis to assess how geopolitical supply disruptions impact the carbon intensity of global gas consumption.

Center for Global Sustainability, University of Maryland

Research Assistant 06/2025 – 08/2025

- Decarbonization Modeling (GCAM): Utilized GCAM to simulate global steel industry transition pathways, explicitly modeling the deployment of hydrogen-based DRI and Electric Arc Furnace (EAF) technologies under net-zero scenarios.

- Circular Economy Integration: Integrated Material Flow Analysis (MFA) frameworks to quantify the potential of scrap steel recycling, assessing how secondary production can bridge the gap between rising material demand and carbon constraints.
- Pathway Analysis: Evaluated the trade-offs between primary and secondary steel production, identifying the impact of circular economy policies and capacity turnover rates on the sector's long-term emission trajectory.

School of Public Policy and Management, Tsinghua University

Research Assistant 03/2024 – 09/2024

- Designed and implemented a database for a sociological study on the environmental practices of catering businesses. Developed the architecture using SQL, optimized indexes, and configured key fields. Created Entity-Relationship Diagrams for efficient data structuring. Handling the import and management of around 35,000 records.
- Performed data cleaning and preprocessing using automation scripts to eliminate duplicates, flag suspicious records, and manage outliers. Leveraged Tableau for visualizing data and creating charts to support preliminary analysis, exploring the link between consumer behavior and corporate environmental policies.

Publications

Wei, T., Sun, L., Sun, J., Vo, S., & Cui, R. Y. (2026). Clean electricity for AI data centers concentrates critical-mineral supply-chain stress. Working paper.

Sun, L., Feng, K., **Wei, T.**, Wang, J., & George, F. (2025). Rethinking LNG in a volatile world: Emissions and strategic lessons from U.S. exports to South Korea, India, and the UK. *SSRN*.

Shen, J., **Wei, T.**, & Cao, C. (2024). Stable discrete segmented reverse diffusion model for solving class imbalance in malicious websites detection. *Proceedings of the IEEE 36th International Conference on Tools with Artificial Intelligence (ICTAI)*.

Cao, C.*, **Wei, T.***, Xu, S., Su, F., & Fang, H. (2023). Comprehensive evaluation of the higher education systems using indicators: Used PCA and EWM methods. *Humanities and Social Sciences Communications*. (**Co-first authors*)

Wei, T., Cao, C., & Shi, Y. (2022). Personal information protection behaviors of consumers in different country contexts and user interface designs. *Proceedings of the 24th International Conference on Human-Computer Interaction (HCII)*.

Ding, X., **Wei, T.**, & Cao, C. (2022). Towards a four-dimensional dynamic trust model in B2C cross-border e-commerce. *Proceedings of the 2022 13th International Conference on E-Education, E-Business, E-Management and E-Learning (IC4E)*.

Internship Experience

Kuaishou Technology: E-commerce Data Operations 03/2023 – 07/2023

- **Industry Data Support:** Leveraged SQL for on-site consumer behavior data in consumer electronics, conducting multidimensional scenario analysis (e.g., search, recommendations, store

pages). Developed specialized business scenarios, abstracted evaluation methods, and empowered similar business scenarios to enhance efficiency in analyzing key focuses.

- **Specialized Business Analysis:** During E-commerce Mall project prep, researched category trends using on-site/off-site classifications, brand features, and user behavior on various devices. Provided data-driven insights through visual reports, supporting and guiding new business model exploration.

Capgemini: IT Consulting Assistant - Product Owner 11/2022 – 02/2023

- Developed and optimized the My BMW and MINI user applications for BMW automobiles using agile methodologies.
- **Requirements Analysis:** Organized and consolidated customer software requirements regarding user experience, focusing on implementable solutions and product prototypes.
- **Feature Design:** Analyzed customer needs, completing feature planning, summaries, and manuals for new software versions. Designed data points within the system, compiled programming requirements for software iterative development, and assisted software engineers in development.
- **Project Management:** Assisted the scrum master in managing the agile process, tracked project progress using Jira and Confluence platforms, and conducted workload evaluations, task breakdowns, and assignments to ensure efficient team collaboration.

Mercer LLC.: Consulting Assistant 02/2022 – 08/2022

- **Industry Research:** Data collection and analysis in power, chemicals, home decor, and new media. Provided insights on industry status and salaries for clients. Compiled the latest HR management theories. Drafted SOPs for customer relationship maintenance.
- **Job Grading:** Compiled and analyzed the client's existing corporate job hierarchy and salary data, and deconstructed the client's human resource management system using Mercer's Global Position Evaluation methodology.

Project Experience

Clean Electricity for AI Data Centers Concentrates Critical-Mineral Supply-Chain Stress 04/2026 – Present

- **First-author manuscript submitted; poster accepted for the AGU Annual Meeting 2026 (Session GC33F, December 2026)**
- **Integrated Modeling Framework:** Designed and coded a framework coupling the GCAM v8.2 integrated assessment model with bottom-up material flow analysis across a 2 × 3 matrix of data center demand and carbon-price scenarios (USA, China, Europe), isolating the marginal power-system expansion, CO_2 emissions and material requirements of data center growth through 2050.
- **Supply-Stress Indices:** Quantified 19 materials across generation, grid-connection and 24/7 carbon-free storage layers with Monte Carlo uncertainty (N = 1,000); developed expansion-absorption and residual-supply indices that test the post-2025 increment against historical-pace, announced-pipeline and maximum-expansion supply trajectories net of background demand.
- **Attribution & Procurement Levers:** Applied Shapley decomposition to separate climate-policy from data center demand effects, quantified the AI share of the exposure, and evaluated eight technology-substitution scenarios plus an integrated material-aware procurement package

(constrained CdTe, silver thrifting, rare-earth-lean wind, LFP storage, renewable-adjacent siting).

Research on the Pattern of Double-circulation Development in Zhejiang Province under Regional Integration – Taking Digital Economy as An Example 05/2021 – 05/2022

- **National College Students' Innovative Entrepreneurial Training Plan Program**
- Developed four theoretical models for different research orientations, conducted experiments and investigations in several countries, and analyzed the data using partial least squares structural equation modeling (PLS-SEM).
- Acknowledged the situation of the transitions and the developments of digital economy among different enterprises under double-circulation strategy, analyzed the connotation and development mode of digital economy.
- Figured out the relationship between RECP and digital economy, explored the changes from RCEP in Asian-Pacific areas.
- Proposed a dynamic model of consumer confidence based on public opinions, customers, companies, and the third-party organizations, studied the trust management and privacy disclosures of consumers in cross-border e-commerce transactions.

Awards & Certifications

- Beta Gamma Sigma - Lifetime Member
- Interdisciplinary Contest in Modeling - Meritorious Winner (2022)
- Interdisciplinary Contest in Modeling - Honorable Mention (2021)